

Technology & Electronics E-Commerce Platform

Project Scope & High-Level Requirements — Version 0.1

PROJECT OVERVIEW

A general Technology & Electronics online store designed to offer products across multiple categories and multiple brands, providing a comprehensive modern e-commerce solution for retail customers and store administrators.

SYSTEM REQUIREMENTS

1. Store Architecture

- General Technology & Electronics Store
- Support for Multiple Categories
- Support for Multiple Brands

2. Product Entity Structure

- Product ID & Name
- Description & Price
- Stock Managed by Admin
- Image
- Category & Brand

3. Product Details

- Display full product information
- Add to Cart functionality

4. Product Discovery

- Keyword Search
- Category & Brand Filtering
- Multi-attribute Sorting

5. Authentication & Authorization

- User Registration & Login
- User Logout
- JWT Authentication Mechanism
- Role-Based Access Control: Customer / Admin Roles

6. Shopping Cart

- Add Product to Cart
- Remove Product from Cart
- Update Item Quantities
- Calculate Total Price automatically

7. Checkout Process

- Order Summary display
- Customer Information entry
- Shipping Information input
- Payment Method selection
- *Note: No real online payment gateway will be used (Simulated/Offline payment)*

8. Order Creation Logic

- Validate Shopping Cart contents
- Validate Stock availability
- Calculate Prices on the Server side
- Create Order and OrderItems records
- Update inventory / Stock counts
- Clear the Customer's Cart
- Execute within a Database Transaction

9. Order Confirmation

- Display Success Message
- Provide generated Order Number
- Display Total Amount
- Set initial Order Status = Pending
- Option to View Order details

10. Admin Order Management

- View all system Orders
- Review Order details
- Confirm Orders
- Cancel Orders
- Update Order Status

11. Order Lifecycle & Status Transitions

- The system must strictly enforce valid status transitions and prevent invalid state transitions.

Pending → Confirmed → Processing → Shipped → Delivered

12. Customer Order Portal

- View "My Orders" history list
- View detailed information for individual orders
- View real-time current Order Status